VULTR — SERVER ONBOARDING SPECIALIST

PHASE 1 MASTER STUDY NOTES | ROMAN URDU

Interview: Thursday, October 15, 2026 | 10:00–10:30 AM Chicago time

Role: Server Onboarding Specialist | Interviewer: Scott Hartman

> Zaroori note: Yeh conceptual workflow job description aur hamari interview practice par based hai, Vultr ka asli internal runbook nahi. Hamesha approved procedure follow karein. PXE, Puppet aur GPU ke concepts seekhna production hands-on experience claim karne ke barabar nahi.

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1. Role aur complete workflow

Server Onboarding Specialist naye racked CPU/GPU server ko hardware handoff se lekar validation aur approved production activation tak prepare karta hai. Is mein BMC, BIOS, firmware, RAID/JBOD, network, PXE imaging, Puppet, health checks, inventory aur documentation shamil hain.

```
1 Hardware handoff: rack / power / cables / inventory
        |
2 iDRAC / BMC: management access / health / logs
        |
3 BIOS + firmware: boot mode / approved versions
        |
4 Storage: RAID / JBOD / disk health
        |
5 Network: NIC -> switch -> VLAN -> DHCP -> PXE services
        |
6 PXE boot / automated Linux installation
        |
```

7 Puppet / Linux health validation
|
8 Inventory / tickets / validation handoff / approval

Linux install ho jana khud-ba-khud production approval nahi hota. Company ka runbook aur sign-off zaroori hai.

2. Step 1 — Hardware handoff

Sab se pehle confirm karein ke sahi server sahi data center, rack aur rack unit mein hai. Serial number, asset tag aur work order match karein. Power feeds, network cables, management cable aur NIC connections remote-hands team se verify karein. Expected CPU/GPU, RAM aur disk count inventory se compare karein. Missing component ya wrong cabling ko record/escalate karein.

Interview English: “First, I would follow the approved onboarding runbook and verify the hardware handoff, including the rack location, power, network connections and inventory details.”

Roman Urdu: “Sab se pehle main approved runbook follow karunga aur rack location, power, cables aur inventory check karunga.”

3. Step 2 — BMC, IPMI, iDRAC aur iLO

BMC (Baseboard Management Controller): Server ka management controller, jo hardware health aur remote power/console ke kaam aata hai.

IPMI (Intelligent Platform Management Interface): Management interface/protocol standard; yeh BMC hardware ka naam nahi.

iDRAC: Dell ka remote management controller.

iLO: HPE/HP ka remote management interface.

Out-of-band (OOB): Host OS down ya uninstalled ho tab bhi separate management path se server access ho sakta hai, agar controller powered aur reachable ho.

Checks: Management IP, authorized access, power state, remote console, sensor readings, event logs, firmware aur hardware alerts. Password ko unsecured ticket ya spreadsheet mein nahi likhna.

Interview English: “Next, I would access Dell iDRAC to verify out-of-band connectivity, hardware health, BIOS settings and firmware versions.”

Failure: iDRAC unreachable ho to management VLAN, IP, cable, switch aur power check karein. Login issue mein approved credentials/vault check karein.

4. Step 3 — BIOS/UEFI aur firmware

BIOS/UEFI server hardware initialize karta hai aur boot behavior control karta hai. Check karein:

- Boot mode UEFI hai ya Legacy, jo approved runbook ke mutabiq ho.

- PXE/network boot enabled hai aur boot order correct hai.

- CPU, RAM, NIC aur storage controller expected tarah detect ho rahe hain.

- BIOS, iDRAC, NIC aur storage-controller firmware approved baseline se match karte hain.

- Hardware warnings ya security settings mein koi issue nahi.

Firmware: Hardware component ko control karne wala low-level software. Har baar latest firmware install karna zaroori nahi; approved version use karna hai. Firmware update se reboot/compatibility issue aa sakta hai; bina approval update na karein.

Interview English: “I would verify BIOS boot settings, detected hardware and the approved BIOS, iDRAC, NIC and storage-controller firmware versions.”

5. Step 4 — RAID, JBOD aur disks

RAID (Redundant Array of Independent Disks): Multiple disks ka logical arrangement, redundancy aur/ya performance ke liye.

- RAID 0: Striping, redundancy nahi; ek disk fail ho to data loss ho sakta hai.

- RAID 1: Mirroring, duplicate data copies.

- RAID 5: Distributed parity, normally ek disk failure tolerate kar sakta hai.

- RAID 10: Mirroring + striping, performance aur redundancy.

JBOD (Just a Bunch Of Disks): Disks generally individually expose hoti hain, traditional RAID array ki tarah combine nahi hotin.

Checks: Disk count, size, health, storage controller, required RAID/JBOD mode, boot disk aur logical volumes. RAID backup ka replacement nahi.

Safety: Disk initialize, RAID create ya mode change karne se data delete ho sakta hai. Approved procedure ke baghair bilkul nahi.

Interview English: “I would verify the required RAID or JBOD configuration and confirm that all expected disks are detected and healthy.”

6. Step 5 — Network verification

Yaad rakhein: NIC -> Switch Port -> VLAN -> DHCP -> PXE Services.

1. NIC: Network Interface Card; physical link aur cable check karein.

2. Switch port: Sahi switch assignment, enabled port aur configuration check karein.

3. VLAN: Virtual LAN; provisioning ke liye correct logical network.

4. DHCP: Dynamic Host Configuration Protocol; IP configuration provide karta hai, aur kuch designs mein boot discovery mein help karta hai.

5. PXE services: TFTP/HTTP ya jo approved network boot service ho, reachable honi chahiye.

IPv4/IPv6: IP addressing ke do protocols; environment ka required addressing/routing design follow karein.

Important: Linux install nahi hua to new server par ip addr nahi chala sakte. iDRAC console, switch status, DHCP logs aur PXE messages use karein.

Interview English: “I would check the NIC link and cabling, verify the switch port and provisioning VLAN, confirm DHCP responds and ensure the PXE boot services are reachable.”

7. Step 6 — PXE boot aur automated imaging

PXE = Preboot Execution Environment. Is se server local OS install hone se pehle network se boot process start karta hai.

```
Server on -> BIOS/UEFI PXE boot -> DHCP IP/boot discovery
        -> Network boot loader (TFTP/HTTP/iPXE)
        -> Linux installer (kernel + initramfs)
        -> Automated Linux install -> reboot -> verify
```

Samajhein: DHCP IP deta hai, Linux install nahi karta. PXE network boot start karta hai; actual automated installation alag image/configuration workflow se hoti hai.

PXE failure troubleshooting:

- Link nahi: NIC, cable, switch, optics.

- IP nahi: VLAN, DHCP scope, DHCP relay aur logs.

- IP mil gaya, boot file nahi: boot server, filename, TFTP/HTTP, firewall.

- Boot loader fail: UEFI/Legacy mismatch ya wrong files.

- Installer fail: image repository, disk configuration, installer logs.

- Install ke baad boot fail: boot order, OS boot loader aur storage.

Interview English: “I would confirm PXE is enabled, check DHCP and boot-file retrieval, monitor automated Linux imaging and validate the first boot. If it fails, I would identify the exact failure stage, review logs and follow the approved troubleshooting process.”

8. Step 7 — Puppet, Ansible aur server validation

Linux install hone ke baad configuration aur health verify karna zaroori hai.

Puppet: Configuration management tool; commonly agent-based aur declarative. Existing approved configuration apply/monitor karne ke liye use hota hai.

Ansible: Aksar SSH-based agentless automation; YAML playbooks use karta hai. Aapka Ansible experience relevant hai, lekin Puppet ka production hands-on claim nahi karna.

Post-install checks: OS/hostname, IP/route/DNS, disks/mounts, packages, services, security settings, monitoring/logging agents, hardware health, Puppet run result.

Puppet fail ho jaye to: Exact error/logs -> hypothesis -> evidence -> root cause -> approved fix/escalation -> authorized rerun -> validation -> documentation. Bina cause samjhe baar baar rerun nahi karna.

```
hostnamectl
ip addr
ip route
lsblk
df -h
```

```
systemctl --failed
sudo journalctl -p err -b
puppet --version
sudo systemctl status puppet
sudo journalctl -u puppet -n 50
# Sirf approval ke saath; configuration change kar sakta hai:
sudo puppet agent -t
```

Interview English: “I would verify Linux health and monitor the approved Puppet configuration workflow. If it failed, I would check the exact errors and logs, gather evidence, follow the approved fix or escalation, rerun only when authorized and validate again.”

9. Step 8 — Documentation aur production handoff

Inventory: Serial number, asset tag, hostname, data center/rack, switch/port, VLAN/IP, BMC management address (password nahi; secure credential reference), CPU/RAM, disks, GPU population, OS/firmware, status, ticket IDs.

Documentation: Imaging result, Puppet report, hardware/network checks, blockers, troubleshooting aur resolutions.

Handoff: Validation team ko complete records aur evidence dein. Required sign-off ke baad hi production-ready mark karein.

DOA = Dead on Arrival: Naya hardware receive hote hi defective.

RMA = Return Merchandise Authorization: Faulty hardware ko vendor se replace/return karne ka process.

Interview English: “I would update inventory and tickets, document validation results and issues, then hand the server over to the validation team. I would only mark it production-ready after approval.”

10. Linux commands, troubleshooting aur safety

Yeh commands installed Linux ya provisioning server par use honge; uninstalled server par nahi.

```
hostnamectl              # Hostname aur OS
ip addr                  # Interfaces / IPs
ip link                  # Link status
ip route                 # Routing
lsblk                    # Disks
df -h                    # Filesystem usage
sudo fdisk -l            # Disk partition info
systemctl --failed       # Failed services
sudo journalctl -p err -b   # Boot errors
sudo journalctl -u dhcpd    # DHCP logs, agar service dhcpd ho
sudo ss -lunp            # UDP listeners
uptime                   # Uptime / load
```

Troubleshooting formula: Symptom confirm -> failure stage -> logs/evidence -> hypothesis -> approved fix/escalation -> revalidate -> document.

Safety: BMC passwords expose na karein; bina approval disk initialize, firmware update, Puppet rerun ya production approval na dein.

11. Mock interview record aur English corrections

Yeh written practice scores hain; speaking performance ya hands-on proficiency ka proof nahi.

- Introduction: 9/10. RHCSA, Linux, Dell/HP, iDRAC/iLO aur enterprise support relevant.

- Steps 1–2: 10/10. Hardware handoff aur iDRAC answer correct.

- Steps 3–4: 7/10. RAID/JBOD mention hua lekin BIOS/firmware miss ho gaya; networking/PXE par jaldi chale gaye.

- Step 5: 9/10. NIC -> switch -> VLAN -> DHCP -> PXE correct. Checklist ko spoken sentences mein convert karein.

- Step 6: 9/10. DHCP, boot files, imaging, logs correct. UEFI/boot-loader detail add karein; example se hat kar khud explain karein.

- Step 7: 5/10. Logs/hypothesis/evidence acha approach tha, lekin OS health aur Puppet workflow miss tha. Original: “I will start with checking logs then hypothesis then evidens the validate and then follow approved procedure.” Correct: “I would start by checking the logs, forming a hypothesis, gathering evidence and validating the findings. Then I would follow the approved troubleshooting procedure.”

- Step 8: 6/10. Inventory sahi, lekin validation evidence aur handoff missing. Original: “i ensure that server info according to inventory system e.g. server serial nos rack location, switch port, IPs, disks details,” Correct: “I would ensure that the server information matches the inventory records, including the serial number, rack location, switch port, IP addresses and disk details.”

Tell me about yourself — English model

“Thank you, Scott. I'm an RHCSA-certified Linux administrator with over six years of experience in Linux administration, production support and troubleshooting. I've supported enterprise environments at American Express, Wells Fargo and AT&T. Earlier in my career, I worked with Dell PowerEdge and HP ProLiant hardware, iDRAC/iLO, RAID, firmware and hardware troubleshooting. I also have Bash and Ansible experience. Vultr interests me because this role combines Linux and bare-metal infrastructure while giving me the opportunity to learn more about large-scale provisioning.”

All eight steps — English interview answer

“I would start with the approved runbook and verify the rack, power, cabling and inventory. Next, I would access iDRAC and check hardware health. Then I would verify BIOS, firmware and RAID or JBOD. I would check NIC link, switch port, VLAN, DHCP and PXE service reachability. After that, I would run the approved PXE/Linux imaging workflow. Once Linux is installed, I would monitor Puppet and validate the OS, network, disks, services and hardware health. Finally, I would update inventory and tickets and hand the server to validation for approved production activation.”

12. Quick revision, glossary aur self-test

Eight words: Hardware -> iDRAC -> BIOS/Firmware -> RAID/JBOD -> Network -> PXE -> Puppet/Validate -> Handoff.

Glossary: NIC = Network Interface Card; BMC = Baseboard Management Controller; IPMI = Intelligent Platform Management Interface; OOB = Out-of-Band; UEFI = Unified Extensible Firmware Interface; RAID = Redundant Array of Independent Disks; JBOD = Just a Bunch Of Disks; VLAN = Virtual LAN; DHCP = Dynamic Host Configuration Protocol; PXE = Preboot Execution Environment; DOA = Dead on Arrival; RMA = Return Merchandise Authorization.

Khud se jawab dein:

1. iDRAC OS ke baghair kaise kaam karta hai?
2. BMC aur IPMI mein kya farq hai?
3. PXE ke liye BIOS/UEFI mein kya verify karna hai?
4. RAID 0, 1, 5, 10 mein kya farq hai?
5. Storage changes ke liye approval kyun chahiye?
6. DHCP se IP na mile to pehle kya check karenge?
7. PXE aur automated Linux installer alag kaise hain?
8. DHCP chale magar boot file fail ho to kya check karenge?
9. Linux install ke baad health checks kya hain?
10. Puppet fail ho to kya approach hogi?
11. Inventory mein kaun si details record hongi?
12. Production-ready status kab milega?

Phase 2 se pehle: Steps 3–4 aur 7 ko dobara apne words mein practice karein. Yeh Phase 1 notes hain; advanced troubleshooting scenarios Phase 2 mein honge.